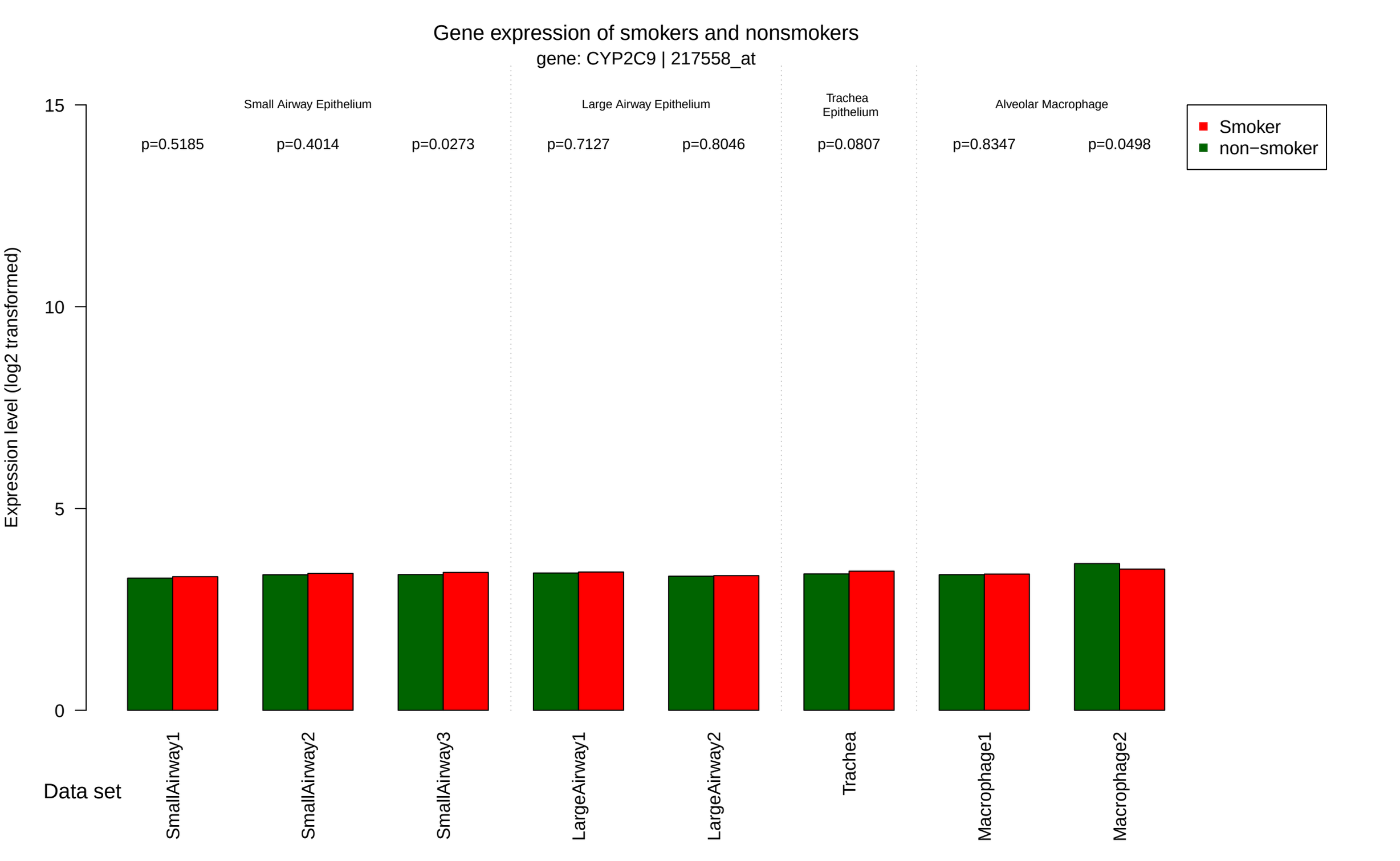
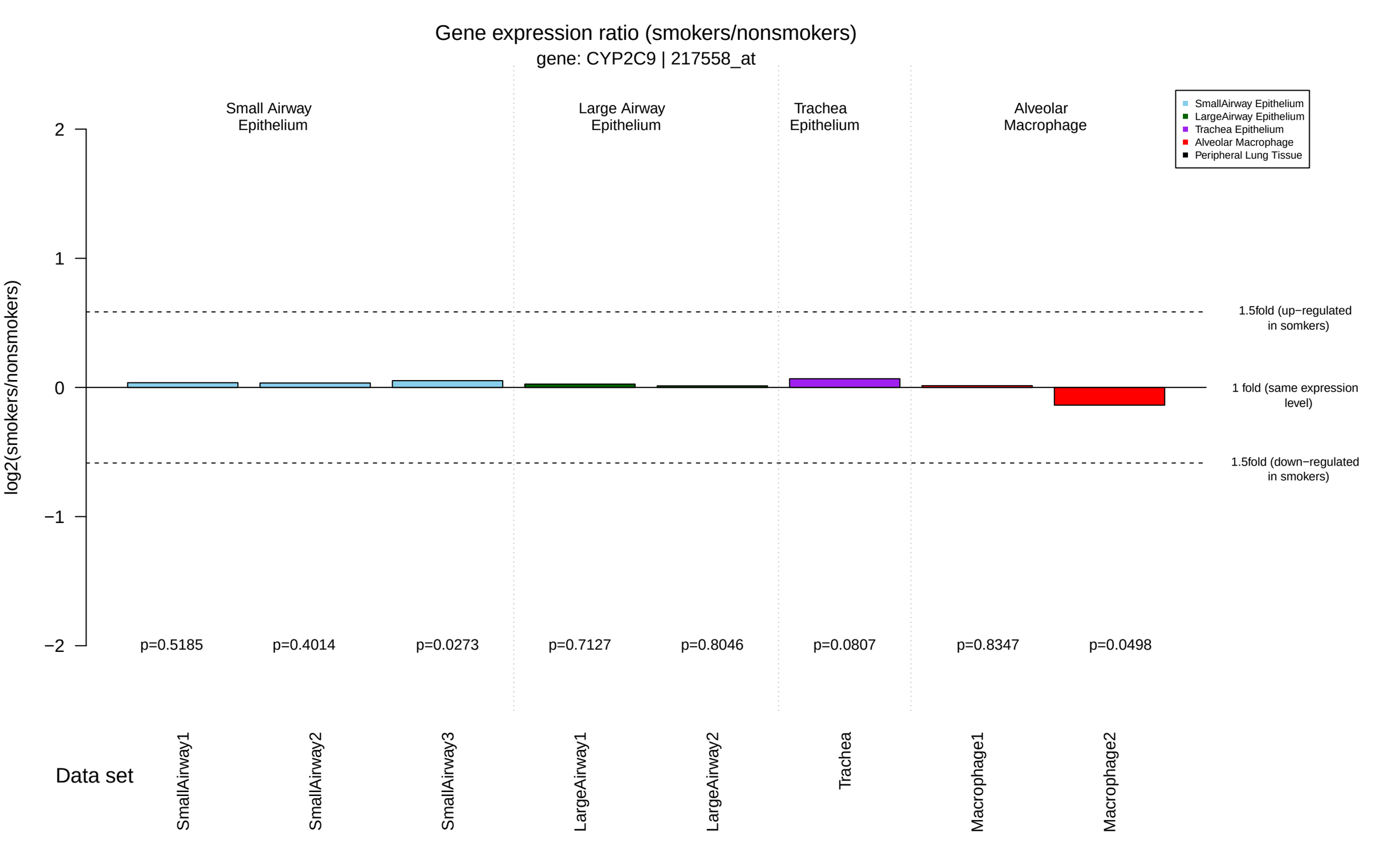
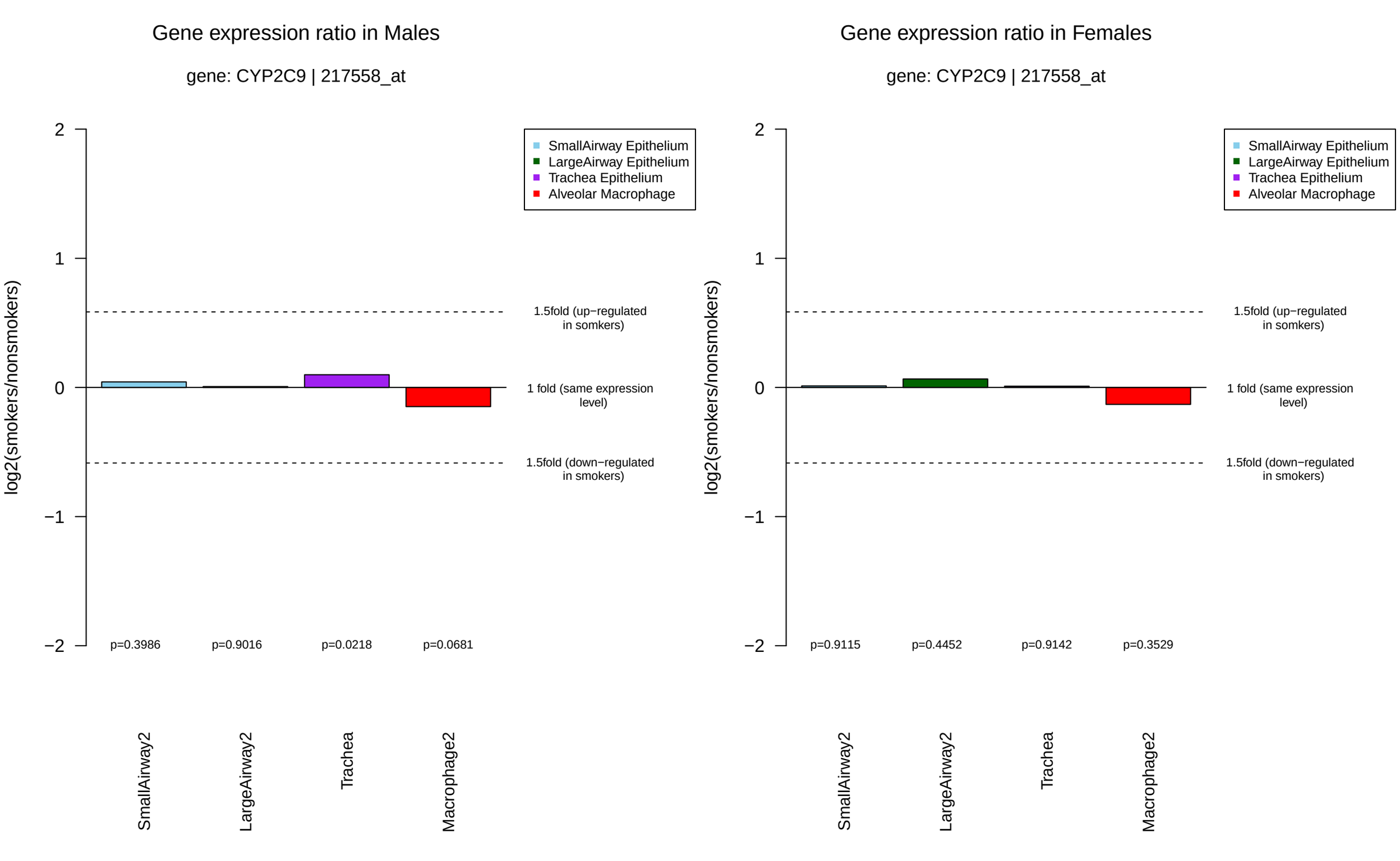
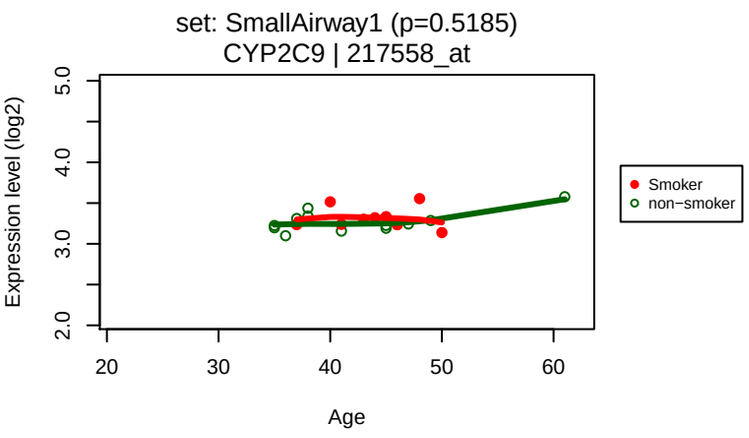
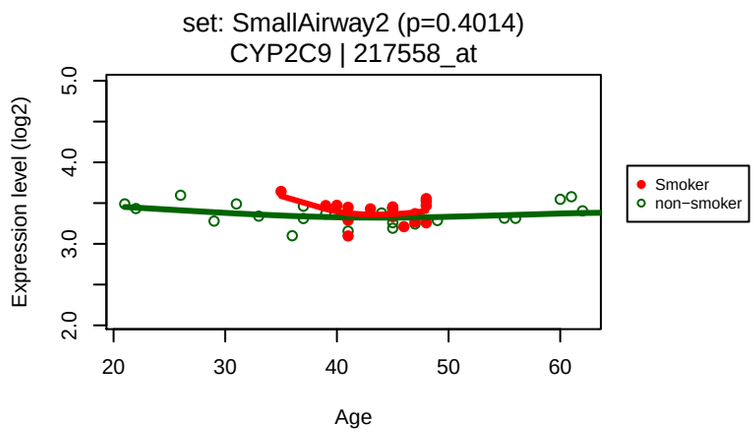
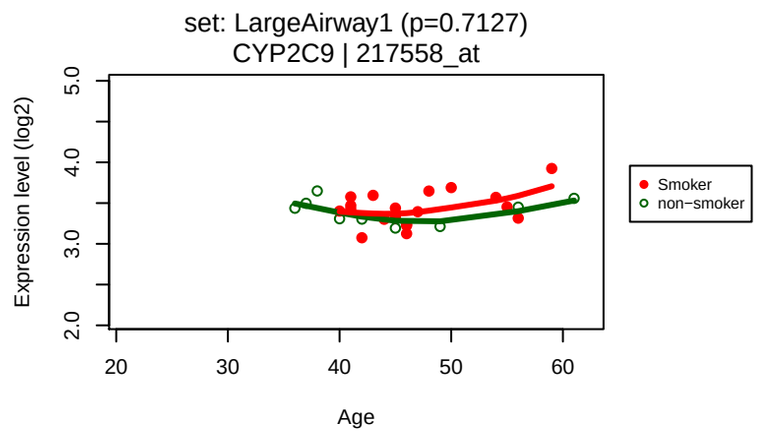

Gene expression of smokers and nonsmokers
gene: CYP2C9 | 217558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5185
p=0.4014
p=0.0273
p=0.7127
p=0.8046
p=0.0807
p=0.8347
p=0.0498
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

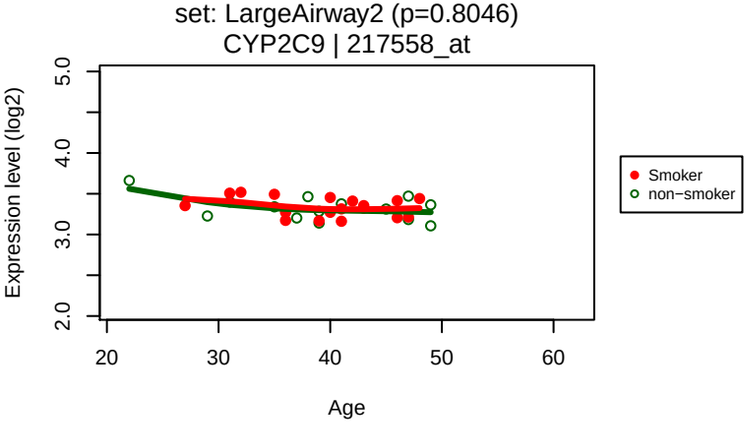

Gene expression ratio (smokers/nonsmokers)
gene: CYP2C9 | 217558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5185
p=0.4014
p=0.0273
p=0.7127
p=0.8046
p=0.0807
p=0.8347
p=0.0498
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

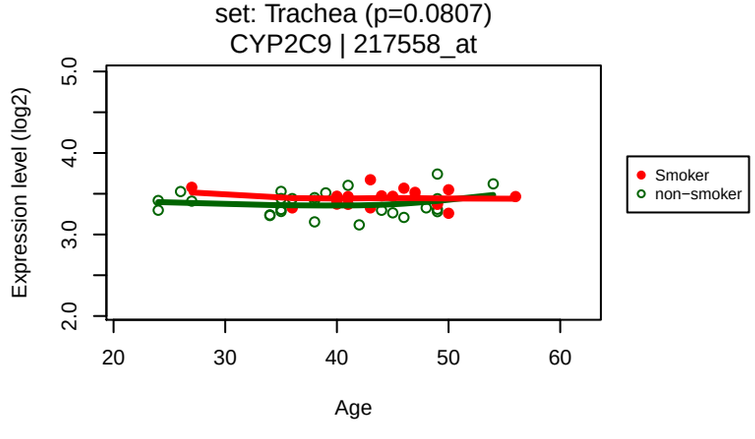

Gene expression ratio in Males
gene: CYP2C9 | 217558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3986
p=0.9016
p=0.0218
p=0.0681
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2C9 | 217558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9115
p=0.4452
p=0.9142
p=0.3529
SmallAirway2
LargeAirway2
Trachea
Macrophage2

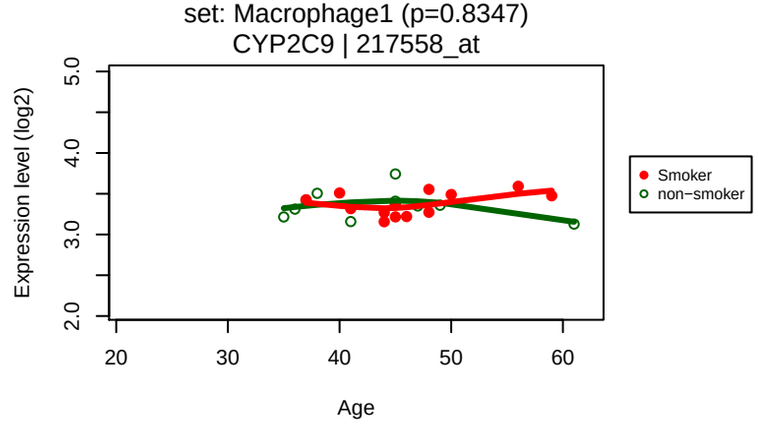

set: SmallAirway1 (p=0.5185)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker

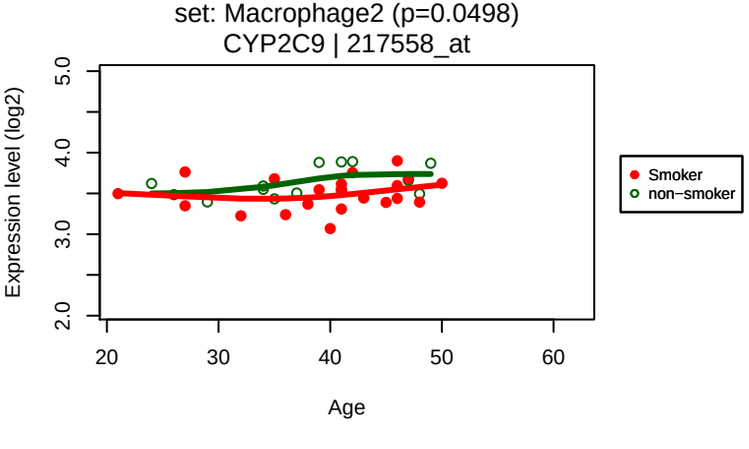

set: SmallAirway2 (p=0.4014)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker

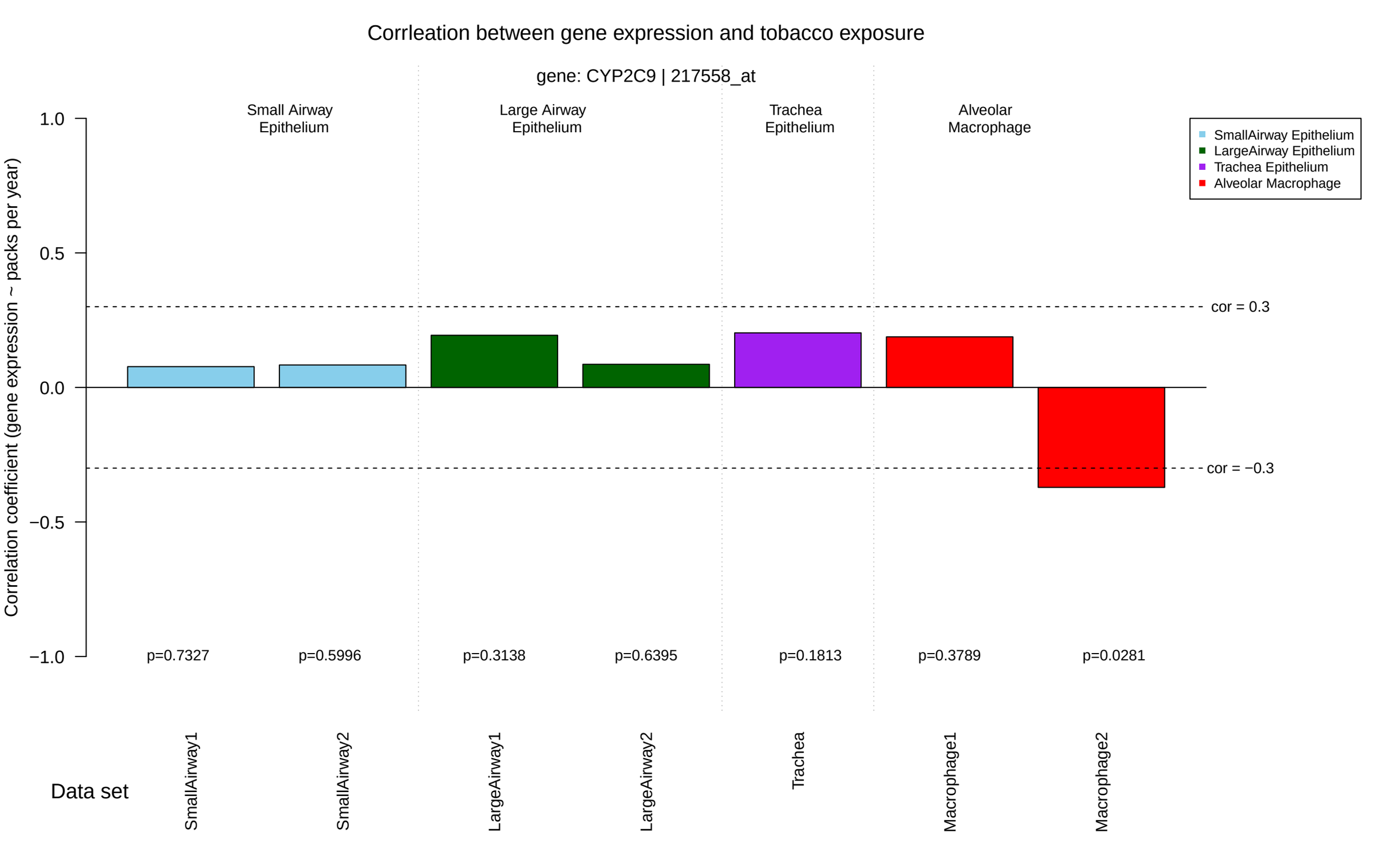

set: LargeAirway1 (p=0.7127)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8046)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0807)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8347)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0498)
CYP2C9 | 217558_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2C9 | 217558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7327
p=0.5996
p=0.3138
p=0.6395
p=0.1813
p=0.3789
p=0.0281
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2